Smart - Waste segregator

Prototyping and Testing (DS3001)

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Introduction

Due to its significant effects on the environment, society, and economy, waste management has drawn more attention in recent years. Ineffective waste management causes climate change, increases pollution, depletes resources, and becomes threats to public health.Developing creative solutions that support sustainable waste management techniques is becoming more and more necessary in response to these issues.

Motivation

- **Environmental Impact:** Waste segregation is essential for lowering pollution levels in the environment and protecting natural resources. We can reduce the damaging consequences of inappropriate waste disposal on ecosystems, water bodies, and air quality by designing a waste segregator prototype.
- **Resource Conservation:**Appropriate waste segregation makes it easier to recycle and reuse materials, which saves energy and precious resources needed to make new goods.
- **Entrepreneurial Opportunities:**A successful trash segregator prototype project may open up new entrepreneurial options for future development and commercialization. It might result in the development of new goods or services that meet market demand for sustainable waste management solutions.

Problem Statement I

- Inefficient waste management procedures lead to environmental degradation, resource depletion, and health risks.
- Despite current recycling activities, a lack of adequate segregation mechanisms leads to mixed waste streams, which prevents recycling operations and worsens degradation of the environment.
- Manual sorting techniques are labor-intensive, time-consuming, and prone to mistake, resulting in low recycling rates and increasing garbage buildup.

Problem Statement II

- The prototype's aim is to increase the efficiency and accuracy of waste segregation, making it much easier to recycle materials and minimise the quantity of garbage sent to garbage dumps.
- As a result -
  Our objective is to design and develop a waste segregator prototype that automates the process of sorting different types of waste, including plastics, paper, metals, and organics.

Existing solutions I

- **Conveyor Belt Systems:**Some waste management facilities make use of conveyor belt systems with human workers to physically separate garbage as it goes along the conveyor. While this is more efficient than wholly manual sorting, it still requires human labour and may not be completely efficient.
- **Eddy Current Separators:**Eddy current separators are used particularly to separate nonferrous metals from waste streams. These devices generate a magnetic field that repels aluminium and other nonferrous metals, separating them from other materials like plastic and paper.

Existing solutions II

- **Automated Sorting Systems:**Advanced waste management facilities have automated sorting systems that use sensors, cameras, and robotic arms to identify and sort various sorts of garbage.
- sorting machines use optical sensors and algorithms to detect and separate waste based on color, shape, and material composition. These techniques are more efficient and accurate than manual methods.

Gaps in the existing solutions I

- **Conveyor Belt Systems:**
  - Still relies on human labor for manual sorting.
  - Efficiency may vary depending on the speed of the conveyor belt and the proficiency of workers.
  - Inadequate for precise sorting requirements for complex waste streams.
- **Eddy Current Separators:**
  - Specialized equipment primarily designed for sorting non-ferrous metals.
  - Not suitable for sorting other types of waste materials such as plastics or paper.

Gaps in the existing solutions II

- **Automated Sorting Systems:**
  - High initial investment and operational costs.
  - Require skilled technicians for maintenance and operation.
  - There is little adaptability to changes in waste composition or sorting requirements.



Our Solution

- Unlike other existing solutions which are difficult to segregate waste or complex to automate and expensive, this is an IoT-based project uses many sensors, including PIR, IR, moisture, proximity, capacitive, and ultrasonic.
- The entire system will be placed so that the garbage bins are used for dumping waste and the bins are separating wastes according to their type.
- The sensors determine the kind of waste that is disposed of in the bin. For example, a moisture sensor detects liquid or moist wastes, a proximity sensor detects metal wastes, a capacitive sensor determines plastic waste and paper waste with varying densities and an infrared sensor determines the bin's maximum capacity. If the bin is full, the system can alert the worker or the supervisor with a "CAUTION" message.



Working Methodology I

Now, let's discuss the most interesting part of the system, the segregation part and how it will operate :

- First, we have to provide power to the hardware, which is loaded with software. When the PIR sensor detects a person, it activates the hardware and displays instructions on the LCD display. If a bin is full, it displays "bins are full, please wait.." Otherwise, it displays "please keep the waste on conveyor belt".
- If an IR sensor detects a dust particle on the conveyor belt, it starts the motor and displays "waste is segregating." Otherwise, the conveyor belt motor will turn off and show the message "Please keep waste on conveyor belt".

Working Methodology II

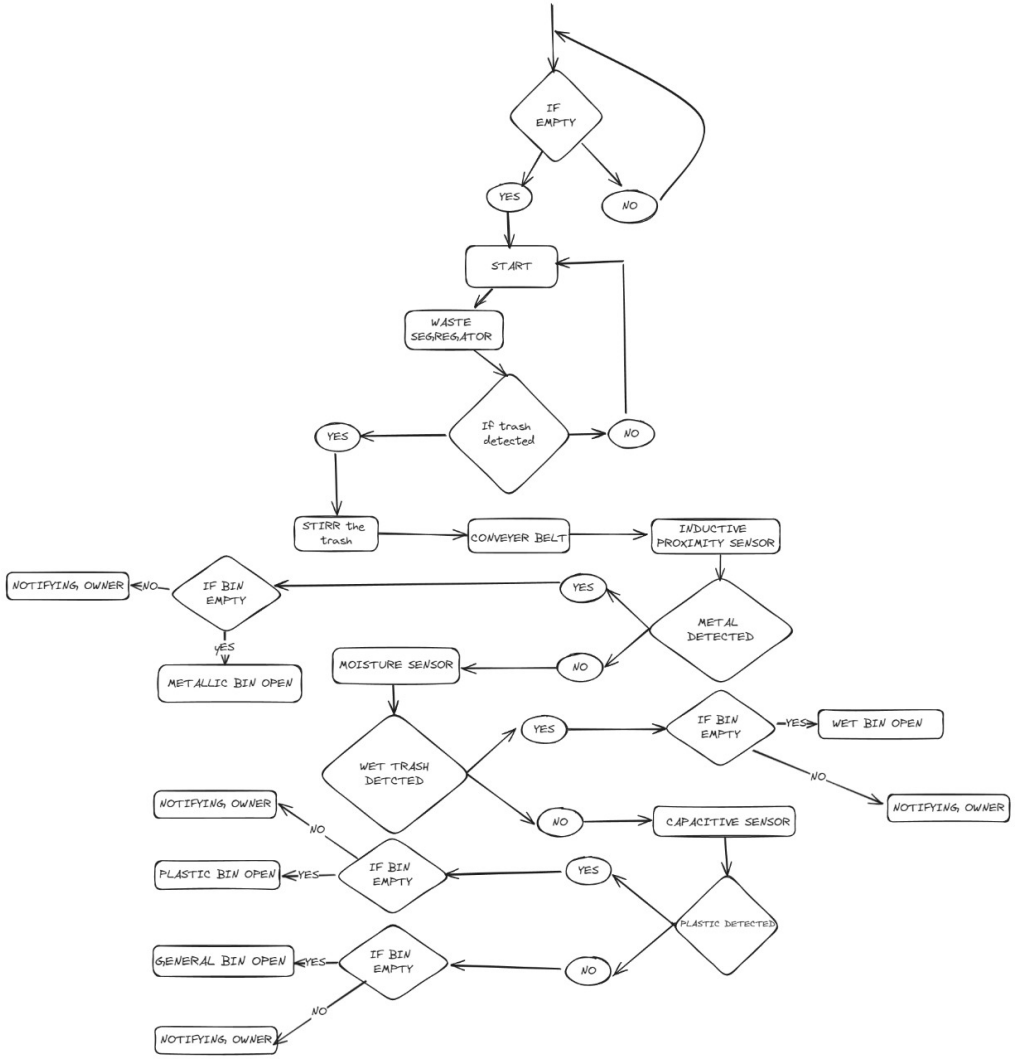
- Sensors are placed on the conveyor belt to detect different types of waste. Four different dust bins are placed at different angles at the end of the conveyor belt, each picking up different waste particles.
- If a metal particle is identified by the inductive proximity sensor, the servo motor rotates the bins to 45deg.The metal particle will be deposited in a specific container at 45deg locations.
- If a wet particle is detected by the moisture sensor, the servo motor rotates the bins to 135deg. Wet particles will be collected in a specific bin located at 135deg.



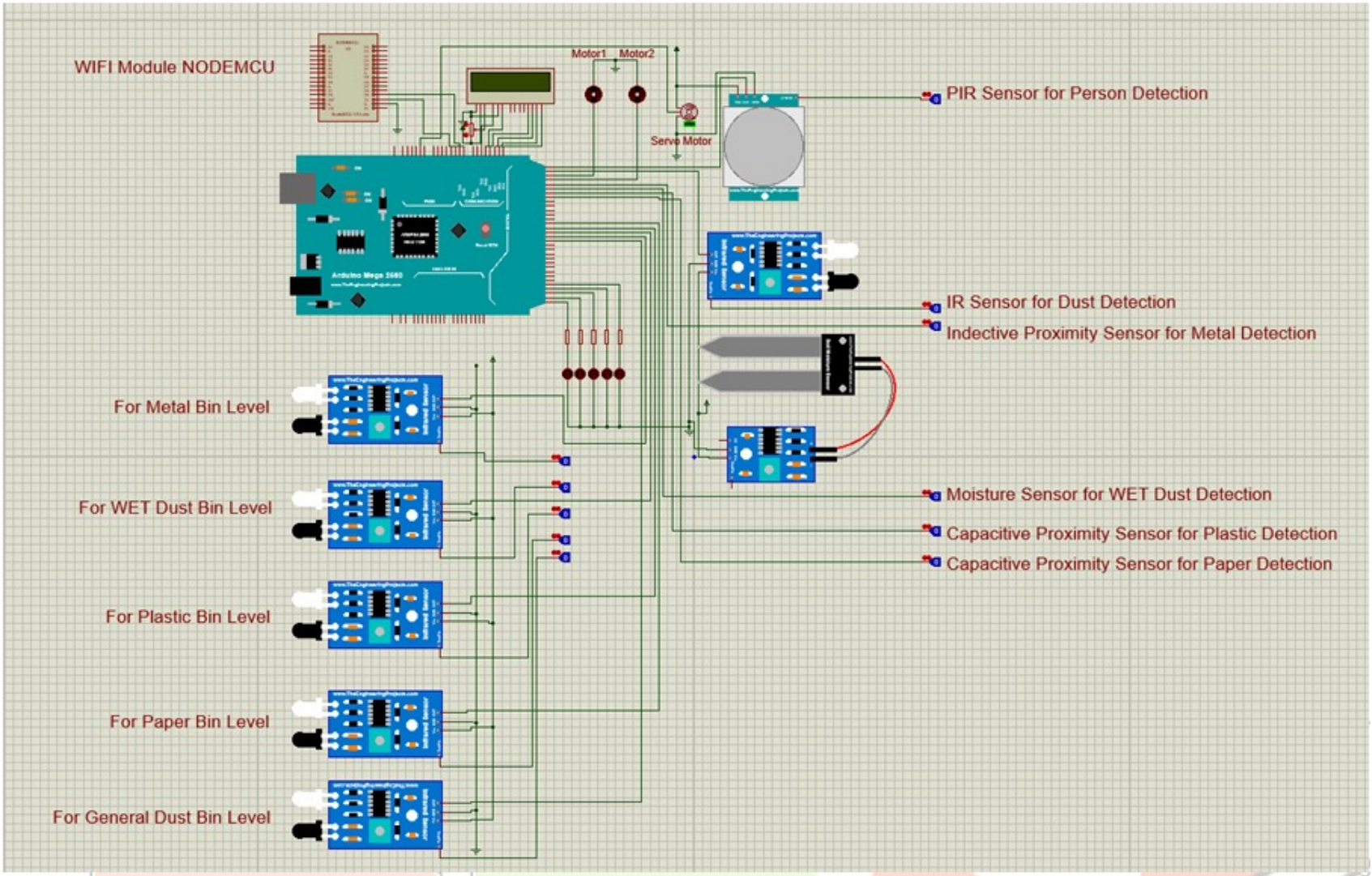
Working Methodology III

- Capacitive sensors detect particles of different densities and thicknesses (10-40 for paper, 60-100 for plastic). The servo motor rotates the plastic bins to 225 and the paper bins to 315.
- If none of the sensors detect a dust particle, the Motor1 turns off and Motor2 turns on. Motor2 will push the dust into the general bin after that, Motor2 will be turned off and the Motor1 will be turned on.
- The IR sensor measures bin levels at the end/start of the process. If any bins are full, an SMS is sent to the respective department to change them. Otherwise, the process will continue....

Flow Chart


IF EMPTY
YES
NO
START
WASTE SEGREGATOR
If trash detected
YES
NO
STIRR the trash
CONVEYER BELT
INDUCTIVE PROXIMITY SENSOR
METAL DETECTED
YES
NO
IF BIN EMPTY
NOTIFYING, OWNER
NO
YES
METALLIC BIN OPEN
MOISTURE SENSOR
WET TRASH DETCTED
YES
IF BIN EMPTY
YES
WET BIN OPEN
NO
NOTIFYING, OWNER
NO
CAPACITIVE SENSOR
PLASTIC DETECTED
YES
IF BIN EMPTY
NOTIFYING, OWNER
NO
PLASTIC BIN OPEN
YES
NO
IF BIN EMPTY
GENERAL BIN OPEN
YES
NO
NOTIFYING, OWNER

Simulation Setup



List of components

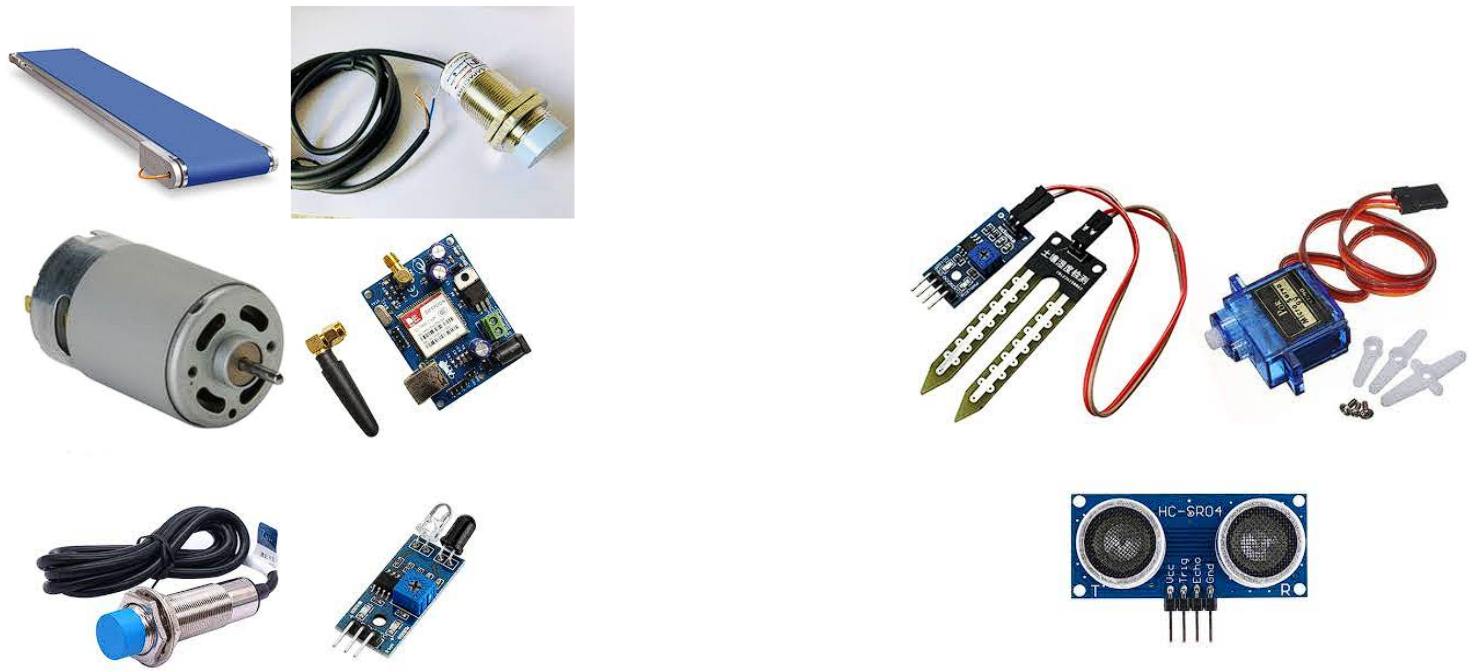


Figure: senors & motors

Cost of components

Component	Quantity	Cost
Inductive proximity sensor	1	Rs.250
Capacitive proximity sensor	1	Rs.530
IR sensor	3	Rs.78
Ultrasonic sensor	1	Rs.99
GSM module	1	Rs.239
moisture sensor	1	Rs.36
DC motor	1	Rs.476
Servo motor	2	Rs.470
Total	11	Rs.2178

Table: Bill Of Materials



Future Scope

- We can improve the accuracy and speed of waste segregation through advancements in sensors.
- Implementing remote monitoring capabilities to allow for real-time monitoring and control of waste segregation processes.
- Integrating the waste segregator into smart city infrastructure for automated waste collection and management.
- We can also use Artificial Intelligence(AI) in near future, for waste segregation which can be more effective and efficient.



Conclusion

In conclusion, the IoT-based smart waste segregator project presents a alternative solution to the growing challenges of waste management.The prototype is based on the development of waste segregator with a low cost and easy to use by anyone.It uses different types of sensors for segregating the waste particles.
The IoT-based smart waste segregator project not only solves the major problem of waste management, but it also represents an innovative rise towards smarter, more sustainable environments.



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Thank You